

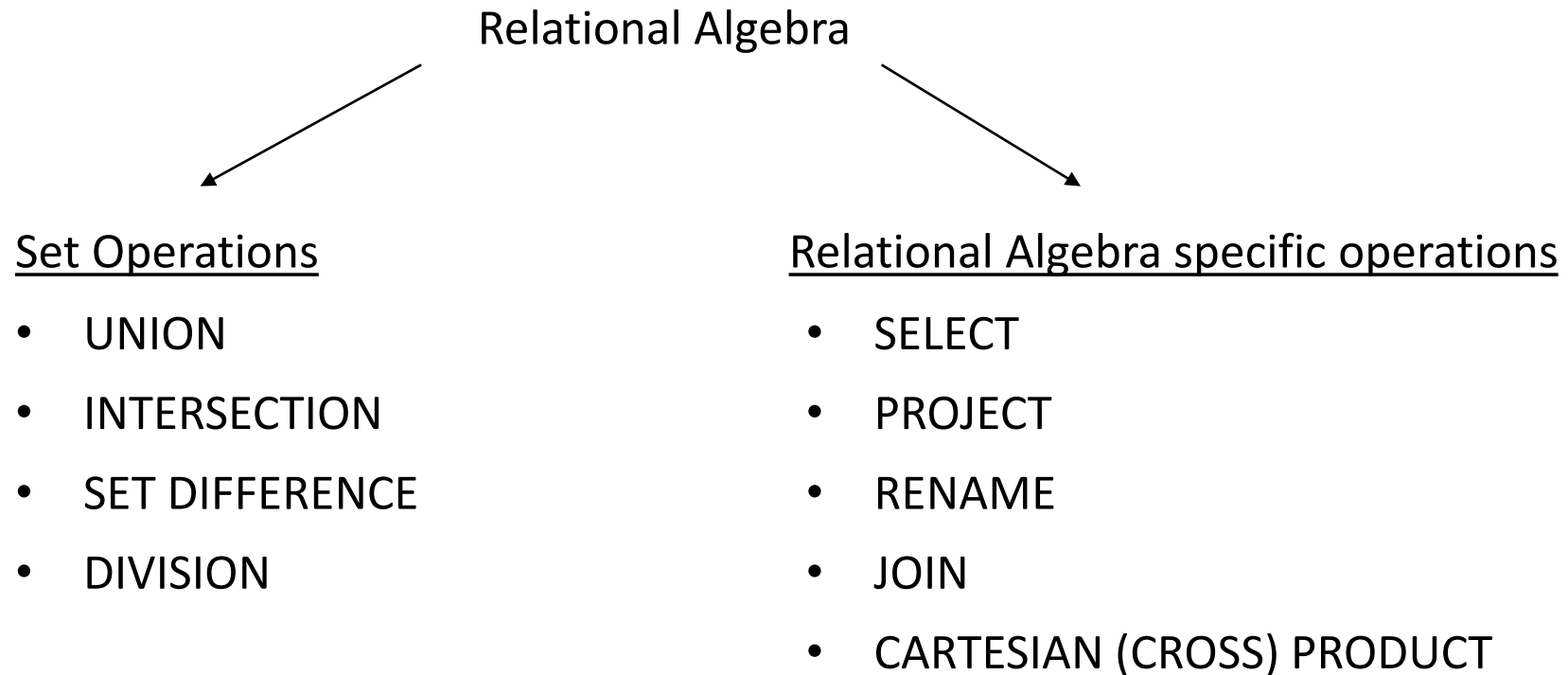
EE4202 Database Systems

RELATIONAL ALGEBRA

Procedural Query Language

INTRODUCTION

- It is a procedural query language
- The basic set of operations for the relational data model is known as *relational algebra*
- Relational algebra is used for implementing and optimizing queries (statements to retrieve data from a database)
- Some of the relational algebraic operations are embedded in SQL.



SELECT

TEACHER
relation

Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Sajitha	Electrical	Software	Senior Lecturer	240,000	3
Kaman	Electrical	Communication	Senior Lecturer	220,000	1
Banjan	Mechanical	Material	Lecturer	146,000	3
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

- Select operation extract tuples from a relation based on meeting a given condition.
- Represented using the sigma (σ) symbol
- $\sigma_{\langle \text{selection condition} \rangle}$ (Relational algebraic expression whose result is a relation)
- Selection condition is a condition applied on attributes.
- Ex: $\sigma_{\text{Area=Communication}}$ (TEACHER)
- Will retrieve all the tuples from TEACHER relation whose Area is equal to Communication.

Name	Department	Area	Designation	Gross Salary	Awards
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Kaman	Electrical	Communication	Senior Lecturer	220,000	1

SELECT

- The **degree** is the number of attributes of the result of relational algebraic operation.
- The degree of a SELECT operation is always equal to degree of the original relation.
- SELECT operation is **commutative**. That is two select operations can be interchanged without affecting the final result.
- Ex: $\sigma_{\text{Designation=Lecturer}}(\sigma_{\text{Awards>5}}(\text{TEACHER})) = \sigma_{\text{Awards>5}}(\sigma_{\text{Designation=Lecturer}}(\text{TEACHER}))$

Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6



Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6

Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6
Banjan	Mechanical	Material	Lecturer	146,000	3
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0



Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6

PROJECT

TEACHER
relation

Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Sajitha	Electrical	Software	Senior Lecturer	240,000	3
Kaman	Electrical	Communication	Senior Lecturer	220,000	1
Banjan	Mechanical	Material	Lecturer	146,000	3
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

$\pi_{\text{Name,Area}}(\text{TEACHER})$

- Project operation extracts specified attributes (columns) of all tuples from a relation
- Represented using the pi (π) symbol
- $\pi_{\langle \text{Set of Attributes} \rangle}$ (Relational algebraic expression whose result is a relation)
- Ex: $\pi_{\text{Name,Area}}(\text{TEACHER})$

Will retrieve all the tuples from TEACHER relation as a new relation with all tuples from the original relation, but having attributes Name and Area only.

- Here, Degree = No. of Attributes of project operation

Name	Area
Dushmantha	Software
Meranda	Communication
Sajitha	Software
Kaman	Communication
Banjan	Material
Paridu	Software
Sithum	Electronics

RENAME

TEACHER
relation

Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Sajitha	Electrical	Software	Senior Lecturer	240,000	3
Kaman	Electrical	Communication	Senior Lecturer	220,000	1
Banjan	Mechanical	Material	Lecturer	146,000	3
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

NEW_RELATION relation

- Rename operation renames a relation or/and attributes of the relation.
- Represented using the rho (ρ) or arrow ($\longleftarrow$) symbol
- Ex: $\rho_{\text{NEW_RELATION}(\text{Teachning_area}, \text{Title})} (\pi_{\text{Area, Designation}}(\text{TEACHER}))$
- or $\text{NEW_RELATION}(\text{Teaching_area}, \text{Title}) \longleftarrow (\pi_{\text{Area, Designation}}(\text{TEACHER}))$

Teaching_area	Title
Software	Lecturer
Communication	Senior Lecturer
Software	Senior Lecturer
Material	Lecturer
Electronics	Lecturer

CARTESIAN PRODUCT

TEACHER

Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6
Meranda	Electrical	Communication	Senior Lecturer	220,000	1

DEPARTMENT

Dept_ID	Head_name	Location	Accreditation
EIE_1	Sajitha	Sector B	Yes
MME_1	Randitha	Sector A	Yes

(TEACHER) X (DEPARTMENT)



Name	Department	Area	Designation	Gross Salary	Awards	Dept_ID	Head_name	Location	Accreditation
Dushmantha	Electrical	Software	Lecturer	146,000	6	EIE_1	Sajitha	Sector B	Yes
Meranda	Electrical	Communication	Senior Lecturer	220,000	1	EIE_1	Sajitha	Sector B	Yes
Dushmantha	Electrical	Software	Lecturer	146,000	6	MME_1	Randitha	Sector A	Yes
Meranda	Electrical	Communication	Senior Lecturer	220,000	1	MME_1	Randitha	Sector A	Yes

- Also known as the cross product or cross join operation.
- Represented using the cross (x)
- Ex: (TEACHER) X (DEPARTMENT)
- If R1 and R2 are relations having t1 and t2 tuples respectively and a1 and a2 amount of attributes, resulting relation consists of t1xt2 tuples and a1+a2 attributes.

INNER JOIN

TEACHER

Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Sajitha	Electrical	Software	Senior Lecturer	240,000	3
Kaman	Electrical	Communication	Senior Lecturer	220,000	1
Banjan	Mechanical	Material	Lecturer	146,000	3
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

DEPARTMENT

Dept_ID	Head_name	Location	Accreditation
EIE_1	Sajitha	Sector B	Yes
MME_1	Randitha	Sector A	Yes
CEE_1	Layanthi	Sector C	Yes



TEACHER ⋈_{Name = Head_name} DEPARTMENT

Name	Department	Area	Designation	Gross Salary	Awards	Dept_ID	Head_name	Location	Accreditation
Sajitha	Electrical	Software	Senior Lecturer	240,000	3	EIE_1	Sajitha	Sector B	Yes

- Combines related tuples from two relations based on a common attribute.
- Represented using the ⋈ symbol.
- Generic form is Relation1 ⋈_{<Join conditions>} Relation2
- Join condition can be (Relation1_Attribute) Θ (Relation2_Attribute) where Θ is a comparison operator (= , !=, > , <, <=, >=).
- If only = comparison operator is used in the join conditions, that join operation is known as an **EQUIJOIN**.
- Equijoin always has two or more attributes whose values are identical.
- Ex: TEACHER ⋈_{Name = Head_name} DEPARTMENT
- NATURAL JOIN (*) eliminates identical attributes by forcing joining relations to have attributes of the same name.

INNER JOIN

TEACHER

Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Sajitha	Electrical	Software	Senior Lecturer	240,000	3
Kaman	Electrical	Communication	Senior Lecturer	220,000	1
Banjan	Mechanical	Material	Lecturer	146,000	3
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

DEPARTMENT

Dept_ID	Head_name	Location	Accreditation
EIE_1	Sajitha	Sector B	Yes
MME_1	Randitha	Sector A	Yes
CEE_1	Layanthi	Sector C	Yes



TEACHER ⋈_{Name = Head_name} DEPARTMENT

Name	Department	Area	Designation	Gross Salary	Awards	Dept_ID	Head_name	Location	Accreditation
Sajitha	Electrical	Software	Senior Lecturer	240,000	3	EIE_1	Sajitha	Sector B	Yes

- A join can be considered as a cartesian product operation followed by a select operation.
- $\text{Relation1} \leftarrow (\text{TEACHER}) \times (\text{DEPARTMENT})$
- $\text{Relation2} \leftarrow \sigma_{\text{Name} = \text{Head_name}}(\text{Relation1})$
- Relation2 is same as the relation obtained from join operation.

SET OPERATIONS

TEACHER

Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Sajitha	Electrical	Software	Senior Lecturer	240,000	3
Kaman	Electrical	Communication	Senior Lecturer	220,000	1
Banjan	Mechanical	Material	Lecturer	146,000	3
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

R1 ← $\sigma_{\text{Area}=\text{Communication}}$ (TEACHER)

Name	Department	Area	Designation	Gross Salary	Awards
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Kaman	Electrical	Communication	Senior Lecturer	220,000	1

R2 ← $\sigma_{\text{Awards} < 3}$ (TEACHER)

Name	Department	Area	Designation	Gross Salary	Awards
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Kaman	Electrical	Communication	Senior Lecturer	220,000	1
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

- Inner UNION, INTERSECTION and MINUS are the basic set operations from the set theory applied in relational algebra.
- These are binary operators meaning that only two relations are involved.
- Two relations must be **type compatible** which means they should have same degree (no. of attributes) and domain of corresponding attributes must be same. Ex: R1 and R2 are type compatible

INNER UNION

$R1 \leftarrow \sigma_{\text{Area}=\text{Communication}}(\text{TEACHER})$

$R2 \leftarrow \sigma_{\text{Awards} < 3}(\text{TEACHER})$

Name	Department	Area	Designation	Gross Salary	Awards	Name	Department	Area	Designation	Gross Salary	Awards
Meranda	Electrical	Communication	Senior Lecturer	220,000	1	Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Kaman	Electrical	Communication	Senior Lecturer	220,000	1	Kaman	Electrical	Communication	Senior Lecturer	220,000	1
						Paridu	Electrical	Software	Lecturer	146,000	1
						Sithum	Electrical	Electronics	Lecturer	146,000	0



$R1 \cup R2$

Name	Department	Area	Designation	Gross Salary	Awards
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Kaman	Electrical	Communication	Senior Lecturer	220,000	1
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

- Inner UNION of type compatible relations A and B is the collection of all tuples that are either in A and B and in both A and B
- Represent by U operator. Resulting relation does not have duplicate tuples
- Ex: $R1 \cup R2$
- **Union operation is commutative.**

INTERSECTION

$R1 \leftarrow \sigma_{\text{Area}=\text{Communication}}(\text{TEACHER})$

Name	Department	Area	Designation	Gross Salary	Awards
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Kaman	Electrical	Communication	Senior Lecturer	220,000	1

$R2 \leftarrow \sigma_{\text{Awards} < 3}(\text{TEACHER})$

Name	Department	Area	Designation	Gross Salary	Awards
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Kaman	Electrical	Communication	Senior Lecturer	220,000	1
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

$R1 \cap R2$



Name	Department	Area	Designation	Gross Salary	Awards
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Kaman	Electrical	Communication	Senior Lecturer	220,000	1

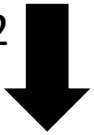
- **INTERSECTION** of type compatible relations A and B is the collection of all tuples that are only in **both** A and B
- Represent by $\cap$ operator. Resulting relation does not have duplicate tuples
- Ex: $R1 \cap R2$
- **Intersection operation is commutative.**

MINUS/SET DIFFERENCE

$R1 \leftarrow \sigma_{\text{Area}=\text{Communication}}(\text{TEACHER})$

Name	Department	Area	Designation	Gross Salary	Awards
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Kaman	Electrical	Communication	Senior Lecturer	220,000	1

$R1 - R2$



Name	Department	Area	Designation	Gross Salary	Awards
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$R2 \leftarrow \sigma_{\text{Awards} < 3}(\text{TEACHER})$

Name	Department	Area	Designation	Gross Salary	Awards
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Kaman	Electrical	Communication	Senior Lecturer	220,000	1
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

$R2 - R1$



Name	Department	Area	Designation	Gross Salary	Awards
Paridu	Electrical	Software	Lecturer	146,000	1
Sithum	Electrical	Electronics	Lecturer	146,000	0

- For type compatible relations A and B, A MINUS B is the set of all tuples in A that are not in B.
- Represent by $-$ operator.
- **Set difference is not commutative.**
- Ex: $(R1 - R2) \neq (R2 - R1)$

DIVISION

R1 (Z)

Name (Y)	Module (X)
Meranda	EE6201
Kaman	EE6201
Dushmantha	EE4202
Sithum	EE3301
Meranda	EE4201
Dushmantha	EE6206
Kaman	EE2301
Sajitha	EE4301
Meranda	EE8201
Sithum	EE4102
Kaman	EE8201

R2 (Y)

Name (Y)
Meranda
Kaman

$$R(X) = R1(Z) \div R2(Y)$$



Module (X)
EE6201
EE8201

- If R1 (Z) and R2 (Y) are relations with attributes Z and Y such that $Z = (X \cup Y)$, then $R1(Z) \div R2(Y)$ produces relation R (X) which contains all tuples t (X) that appear in R1 (Z) in combination with every tuple of R2 (Y).
- Represent by $\div$ operator.
- Not commutative
- Ex: $R(X) = R1(Z) \div R2(Y)$

RELATIONAL ALGEBRA - Extensions

OUTER JOIN

TEACHER

Name	Department	Area	Designation	Gross Salary	Awards
Dushmantha	Electrical	Software	Lecturer	146,000	6
Meranda	Electrical	Communication	Senior Lecturer	220,000	1
Sajitha	Electrical	Software	Senior Lecturer	240,000	3

DEPARTMENT

Dept_ID	Head_name	Location	Accreditation
EIE_1	Sajitha	Sector B	Yes
MME_1	Randitha	Sector A	Yes
CEE_1	Layanthi	Sector C	Yes

TEACHER ⋈_{Name = Head_name} DEPARTMENT

Name	Department	Area	Designation	Gross Salary	Awards	Dept_ID	Head_name	Location	Accreditation
Dushmantha	Electrical	Software	Lecturer	146,000	6	NULL	NULL	NULL	NULL
Meranda	Electrical	Communication	Senior Lecturer	220,000	1	NULL	NULL	NULL	NULL
Sajitha	Electrical	Software	Senior Lecturer	240,000	3	EIE_1	Sajitha	Sector B	Yes

TEACHER ⋈_{Name = Head_name} DEPARTMENT

Name	Department	Area	Designation	Gross Salary	Awards	Dept_ID	Head_name	Location	Accreditation
Sajitha	Electrical	Software	Senior Lecturer	240,000	3	EIE_1	Sajitha	Sector B	Yes
NULL	NULL	NULL	NULL	NULL	NULL	MME_1	Randitha	Sector A	Yes
NULL	NULL	NULL	NULL	NULL	NULL	CEE_1	Layanthi	Sector C	Yes

- Natural Join and Θ join are all inner joins which only joins tuples with matching condition.
- In outer join, tuples from one or both relations appear in the result which do not meet the join condition.
- Outer join is named according to the relation in which all tuples are kept.
- Left outer join $\bowtie_{\leftarrow}$ keeps all tuples of the left/first relation regardless of join condition. Unmatched tuples in the second/right relation is padded with nulls. Same is true for right outer join $\bowtie_{\rightarrow}$
- In full outer join, $\bowtie_{\leftarrow\rightarrow}$ all tuples from left and right relations are kept. If no matching condition is found, the corresponding entries will be padded with null values.

OUTER JOIN

TEACHER(R1)

Name	Department	Area	Designation
Dushmantha	Electrical	Software	Lecturer
Meranda	Electrical	Communication	Senior Lecturer
Sajitha	Electrical	Software	Senior Lecturer

MANAGER(R2)

Name	Department	Head_ID	Experience
Sajitha	Electrical	H005	16
Ranjitha	Mechanical	H002	18
Layanthi	Civil	H003	13

R1 outer union R2

Name	Department	Area	Designation	Hed_ID	Experience
Dushmantha	Electrical	Software	Lecturer	NULL	NULL
Meranda	Electrical	Communication	Senior Lecturer	NULL	NULL
Sajitha	Electrical	Software	Senior Lecturer	H005	16
Randitha	Mechanical	NULL	NULL	H002	18
Layanthi	Civil	NULL	NULL	H003	13

- Inner UNION is defined for type compatible relations while outer UNION is defined for relations consisting of some common attributes but not type compatible. i.e. partially compatible
- Two tuples in partially type compatible relations are said to **match** if their common attributes are equal and will have values for all its attributes.
- In outer union result, matching tuples appear only once.